

Questions with Answer Keys

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Q1: 16 March (Shift 1) - Single Correct

Let $S_k = \sum_{r=1}^k \tan^{-1}\left(\frac{6^r}{2^{2r+1} + 3^{2r+1}}\right)$. Then $\lim_{k \rightarrow \infty} S_k$ equal to :

- (1) $\tan^{-1}\left(\frac{3}{2}\right)$
- (2) $\frac{\pi}{2}$
- (3) $\cot^{-1}\left(\frac{3}{2}\right)$
- (4) $\tan^{-1}(3)$

Q2: 16 March (Shift 2) - Single Correct

Given that the inverse trigonometric functions take principal values only. Then, the number of real values of x which satisfy

$\sin^{-1}\left(\frac{3x}{5}\right) + \sin^{-1}\left(\frac{4x}{5}\right) = \sin^{-1} x$ is equal to:

- (1) 2
- (2) 1
- (3) 3
- (4) 0

Q3: 17 March (Shift 1) - Single Correct

If $\cot^{-1}(\alpha) = \cot^{-1} 2 + \cot^{-1} 8 + \cot^{-1} 18 + \cot^{-1} 32 + \dots$ upto 100 terms, then α is :

- (1) 1.01
- (2) 1
- (3) 1.02
- (4) 1.03

Q4: 17 March (Shift 1) - Single Correct

The sum of possible values of x for $\tan^{-1}(x+1) + \cot^{-1}\left(\frac{1}{x-1}\right) = \tan^{-1}\left(\frac{8}{31}\right)$ is

- (1) $-\frac{32}{4}$

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- (2) $-\frac{31}{4}$
- (3) $-\frac{30}{4}$
- (4) $-\frac{33}{4}$

Q5: 17 March (Shift 2) - Single Correct

The number of solutions of the equation $\sin^{-1}\left[x^2 + \frac{1}{3}\right] + \cos^{-1}\left[x^2 - \frac{2}{3}\right] = x^2$ for $x \in [-1, 1]$, and $[x]$ denotes the greatest integer less than or equal to x , is :

- (1) 2
- (2) 0
- (3) 4
- (4) Infinite

Answer Key**Q1 (3)****Q2 (3)****Q3 (1)****Q4 (1)****Q5 (2)**

Q1 (3)

$$S_k = \sum_{r=1}^k \tan^{-1} \left(\frac{6^r}{2^{2r+1} + 3^{2r+1}} \right)$$

Divide by 3^{2x}

$$\sum_{r=1}^k \tan^{-1} \left(\frac{\left(\frac{2}{3}\right)^r}{\left(\frac{2}{3}\right)^{2r} \cdot 2 + 3} \right)$$

$$\sum_{r=1}^k \tan^{-1} \left(\frac{\left(\frac{2}{3}\right)^r}{3 \left(\left(\frac{2}{3}\right)^{2r+1} + 1 \right)} \right)$$

$$\text{Let } \left(\frac{2}{3}\right)^r = t$$

$$\sum_{r=1}^k \tan^{-1} \left(\frac{\frac{t}{3}}{1 + \frac{2}{3}t^2} \right)$$

$$\sum_{r=1}^k \tan^{-1} \left(\frac{t - \frac{2t}{3}}{1 + t \cdot \frac{2t}{3}} \right)$$

$$\sum_{r=1}^k \left(\tan^{-1}(t) - \tan^{-1}\left(\frac{2t}{3}\right) \right)$$

$$\sum_{r=1}^k \left(\tan^{-1} \left(\frac{2}{3} \right)^r - \tan^{-1} \left(\frac{2}{3} \right)^{r+1} \right)$$

$$S_k = \tan^{-1} \left(\frac{2}{3} \right) - \tan^{-1} \left(\frac{2}{3} \right)^{k+1}$$

$$S_\infty = \lim_{k \rightarrow \infty} \left(\tan^{-1} \left(\frac{2}{3} \right) - \tan^{-1} \left(\frac{2}{3} \right)^{k+1} \right)$$

$$= \tan^{-1} \left(\frac{2}{3} \right) - \tan^{-1}(0)$$

$$\therefore S_\infty = \tan^{-1} \left(\frac{2}{3} \right) = \cot^{-1} \left(\frac{3}{2} \right)$$

Q2 (3)

$$\sin^{-1} \frac{3x}{5} + \sin^{-1} \frac{4x}{5} = \sin^{-1} x$$

$$\sin^{-1} \left(\frac{3x}{5} \sqrt{1 - \frac{16x^2}{25}} + \frac{4x}{5} \sqrt{1 - \frac{9x^2}{25}} \right) = \sin^{-1} x$$

$$\frac{3x}{5} \sqrt{1 - \frac{16x^2}{25}} + \frac{4x}{5} \sqrt{1 - \frac{9x^2}{25}} = x$$

$$x = 0, 3\sqrt{25 - 16x^2} + 4\sqrt{25 - 9x^2} = 25$$

$$4\sqrt{25 - 9x^2} = 25 - 3\sqrt{25 - 16x^2} \text{ squaring we get}$$

$$16(25 - 9x^2) = 625 + 9(25 - 16x^2) - 150\sqrt{25 - 16x^2}$$

$$400 = 625 + 225 - 150\sqrt{25 - 16x^2}$$

$$\sqrt{25 - 16x^2} = 3 \Rightarrow 25 - 16x^2 = 9$$

$$\Rightarrow x^2 = 1$$

Hints and Solutions

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Put $x = 0, 1, -1$ in the original equation

We see that all values satisfy the original equation.

Number of solution = 3

Q3 (1)

$$\cot^{-1}(\alpha) = \cot^{-1}(2) + \cot^{-1}(8) + \cot^{-1}(18) + \dots$$

$$= \sum_{n=1}^{100} \tan^{-1}\left(\frac{2}{4n^2}\right)$$

$$= \sum_{n=1}^{100} \tan^{-1}\left(\frac{(2n+1)-(2n-1)}{1+(2n+1)(2n-1)}\right)$$

$$= \sum_{n=1}^{100} \tan^{-1}(2n+1) - \tan^{-1}(2n-1)$$

$$= \tan^{-1} 201 - \tan^{-1} 1$$

$$= \tan^{-1}\left(\frac{200}{202}\right)$$

$$\therefore \cot^{-1}(\alpha) = \cot^{-1}\left(\frac{202}{200}\right)$$

$$\alpha = 1.01$$

Q4 (1)

$$\tan^{-1}(x+1) + \cot^{-1}\left(\frac{1}{x-1}\right) = \tan^{-1} \frac{8}{31}$$

Taking tangent both sides :-

$$\frac{(x+1)+(x-1)}{1-(x^2-1)} = \frac{8}{31}$$

$$\Rightarrow \frac{2x}{2-x^2} = \frac{8}{31}$$

$$\Rightarrow 4x^2 + 31x - 8 = 0$$

$$\Rightarrow x = -8, \frac{1}{4}$$

But, if $x = \frac{1}{4}$

$$\tan^{-1}(x+1) \in \left(0, \frac{\pi}{2}\right)$$

$$\& \cot^{-1}\left(\frac{1}{x-1}\right) \in \left(\frac{\pi}{2}, \pi\right)$$

$$\Rightarrow \text{LHS} > \frac{\pi}{2} \& \text{RHS} < \frac{\pi}{2}$$

(Not possible)

Hence, $x = -8$

Hints and Solutions

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Q5 (2)

Given equation

$$\sin^{-1}\left[x^2 + \frac{1}{3}\right] + \cos^{-1}\left[x^2 - \frac{2}{3}\right] = x^2$$

Now, $\sin^{-1}\left[x^2 + \frac{1}{3}\right]$ is defined if

$$-1 \leq x^2 + \frac{1}{3} < 2 \Rightarrow \frac{-4}{3} \leq x^2 < \frac{5}{3}$$

$$\Rightarrow 0 \leq x^2 < \frac{5}{3}$$

and $\cos^{-1}\left[x^2 - \frac{2}{3}\right]$ is defined if

$$-1 \leq x^2 - \frac{2}{3} < 2 \Rightarrow \frac{-1}{3} \leq x^2 < \frac{8}{3}$$

$$\Rightarrow 0 \leq x^2 < \frac{8}{3}$$

So, from (1) and (2) we can conclude

$$0 \leq x^2 < \frac{5}{3}$$

Case - I if $0 \leq x^2 < \frac{2}{3}$

$$\sin^{-1}(0) + \cos^{-1}(-1) = x^2$$

$$\Rightarrow x + \pi = x^2$$

$$\Rightarrow x^2 = \pi$$

$$\text{but } \pi \notin \left[0, \frac{2}{3}\right)$$

 $\Rightarrow$ No value of 'x'Case - II if $\frac{2}{3} \leq x^2 < \frac{5}{3}$

$$\sin^{-1}(1) + \cos^{-1}(0) = x^2$$

$$\Rightarrow \frac{\pi}{2} + \frac{\pi}{2} = x^2$$

$$\Rightarrow x^2 = \pi$$

$$\text{but } \pi \notin \left[\frac{2}{3}, \frac{5}{3}\right)$$

 $\Rightarrow$ No value of 'x'

So, number of solutions of the equation is zero.